

House Price Prediction

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1. Introduction

This analytical report explores the Ames Housing dataset to understand how various structural, locational, and categorical features influence residential property values. The project evaluates the role of zoning, lot configuration, basement presence, overall condition, remodeling year, exterior materials, and numerical distributions in shaping house prices. The document presents a structured, end-to-end analysis from data preparation to insight generation, backed by charts and feature engineering, culminating in a clear understanding of the strongest value drivers in the dataset.

2. Methodology

The methodology follows a systematic data-driven approach starting with dataset ingestion using Python's pandas library. An initial structural exploration was completed through head samples, info summaries, and descriptive statistics. Data cleaning involved imputing missing numeric values with mean and categorical values with mode, ensuring a consistent and analysis-ready dataset.

A full Exploratory Data Analysis (EDA) was conducted using scatterplots, bar charts, heatmaps, histograms, and KDE curves. These visuals helped identify distribution patterns, outliers, correlations, and categorical value patterns. Feature engineering expanded the dataset by generating age-based variables, log-transformed targets, scaled continuous fields, and one-hot encoded categorical columns. Business-specific analytical questions were then answered using grouped statistics and visual evidence.

This methodology ensures the findings are both statistically sound and business-relevant.

3. Requirement Analysis

The core requirements of this project revolve around answering specific business questions related to property valuation. Stakeholders need clarity on how zoning, lot configuration, remodeling activity, basement size, and house condition impact pricing. The requirement also includes identifying strong predictive features to support future machine learning models. Visualization was another major requirement to convert raw numerical signals into intuitive business narratives. The dataset needed imputation, scaling, and encoding to support modeling — especially for algorithms like Adaboost and XGBoost. The objective was not only to explain price behavior but also to build a foundation for reliable price prediction.

4. Other Parameters & Dataset Characteristics

The dataset shows skewed distributions in major numeric features like LotArea, TotalBsmtSF, and SalePrice. Log transformations successfully improved normality and reduced variance. Correlation analysis revealed that no single numerical feature has a very strong relationship with SalePrice, but basement area and year-based variables demonstrated meaningful positive relationships. Categorical variables such as zoning and lot configuration showed clear pricing tiers, proving location-based variables offer major pricing influence even when correlation coefficients do not fully capture them.

Exterior material (Exterior1st) also displayed strong pricing variation, with Vinyl Siding properties outperforming several other categories. Outliers in SalePrice were present but expected, representing genuine premium-market homes.

5. Visualizations Used in the Analysis

A complete set of visuals supports the analysis, including:

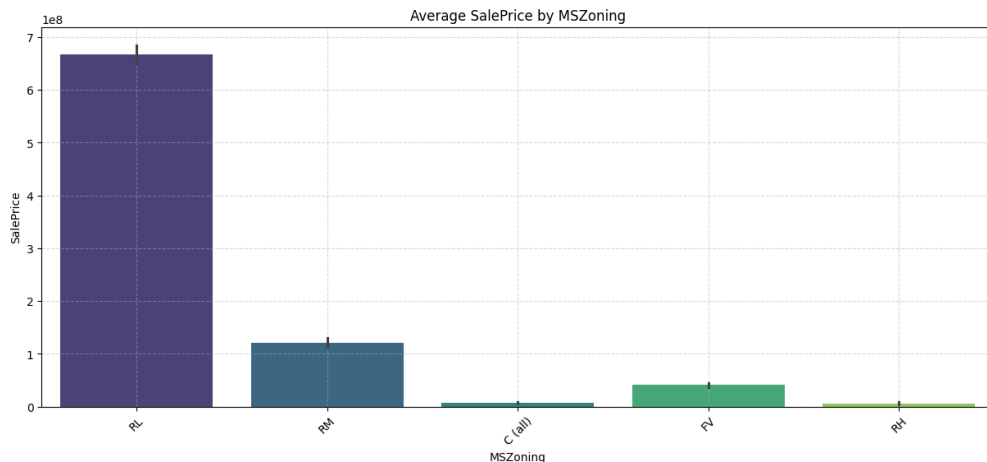
- Scatterplots: Basement size vs SalePrice

- Barplots: MSZoning, LotConfig, OverallCond, Exterior1st
- Lineplots: YearBuilt vs SalePrice, YearRemodAdd vs SalePrice
- Histograms & KDE curves: SalePrice, LotArea, TotalBsmtSF (before and after log-transform)
- Heatmaps: Correlation analysis
- Boxplots: Condition vs SalePrice

6. Insights from Charts & Analysis

Zoning Influence:

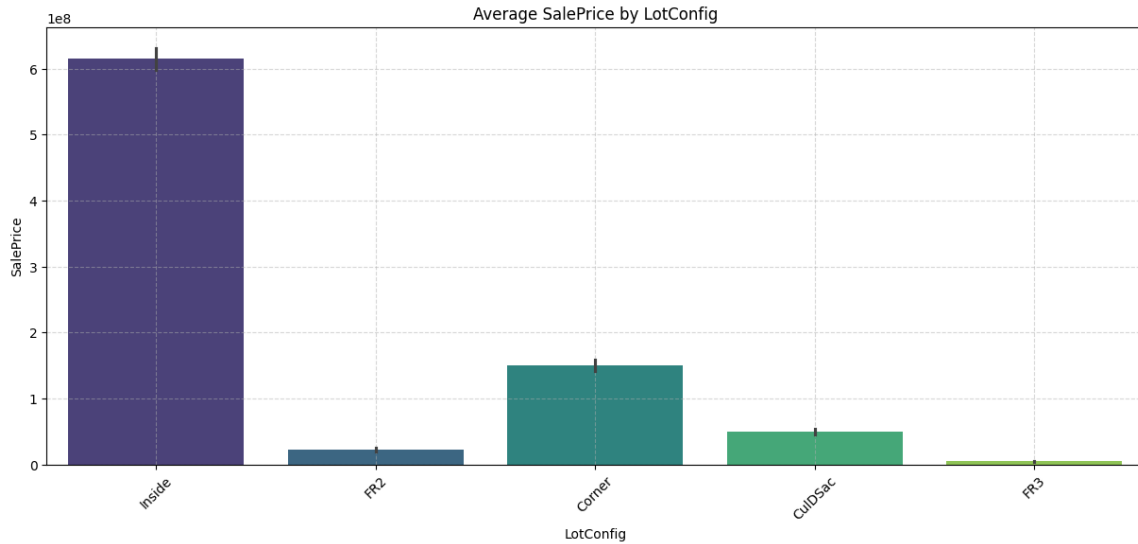
Zoning plays a critical role in shaping property values. RL and FV consistently sit at the top of the pricing ladder due to better neighborhood planning, larger lots, and lower congestion. RM and RH, representing medium- and high-density areas, show suppressed pricing. This behavior highlights zoning as a powerful proxy for buyer preference, neighborhood quality, and long-term appreciation potential.



Lot Configuration Insights:

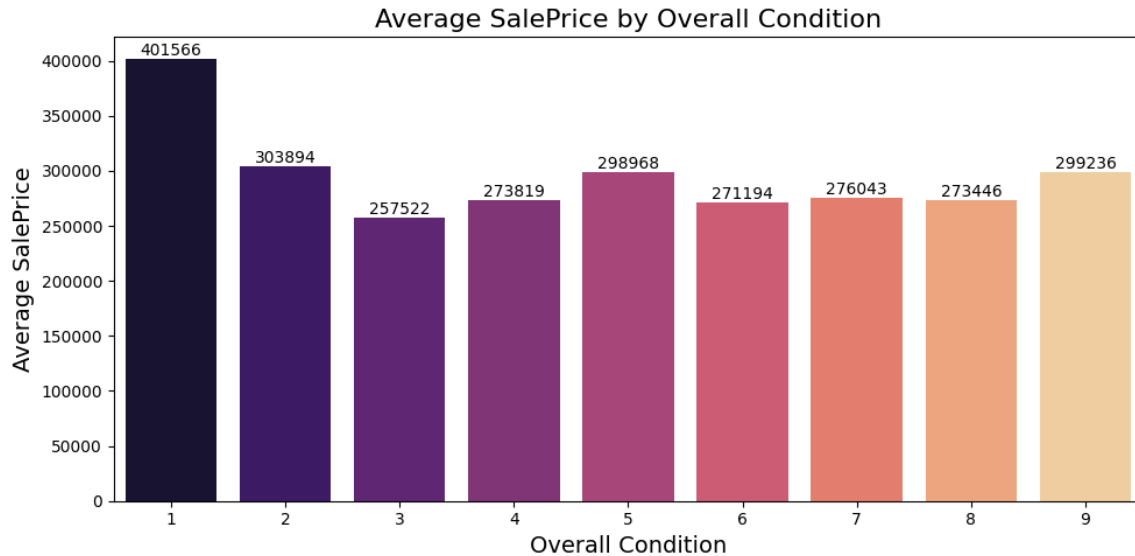
The lot configuration visuals show a clear value hierarchy. FR3 and Corner lots command price premiums because they offer better street exposure, larger frontage, and overall appeal. Inside lots remain predictable but lack

differentiation, while FR2 lots remain the lowest because of narrow frontage and limited structural flexibility. A front-facing advantage directly converts into monetary advantage.



Overall Condition Insights:

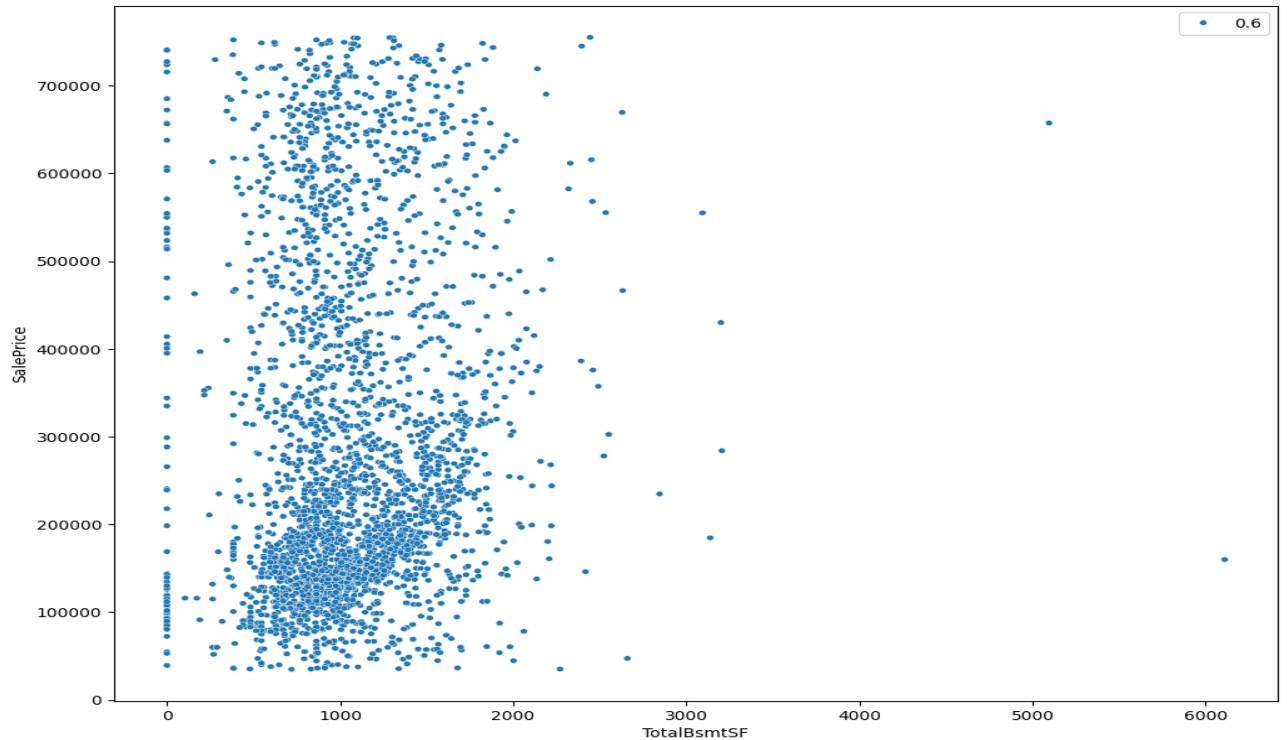
While overall condition intuitively feels like a strong driver, the data indicates otherwise. The average prices across conditions 3–8 remain fairly stable with minimal variation. Extreme categories, like condition 1 and 9, show high averages, but these anomalies are tied to land value, not the condition itself. The trend confirms that buyers reward structural size, location, and upgrades more than basic condition ratings. Condition is only a secondary pricing signal.



Basement Area Influence:

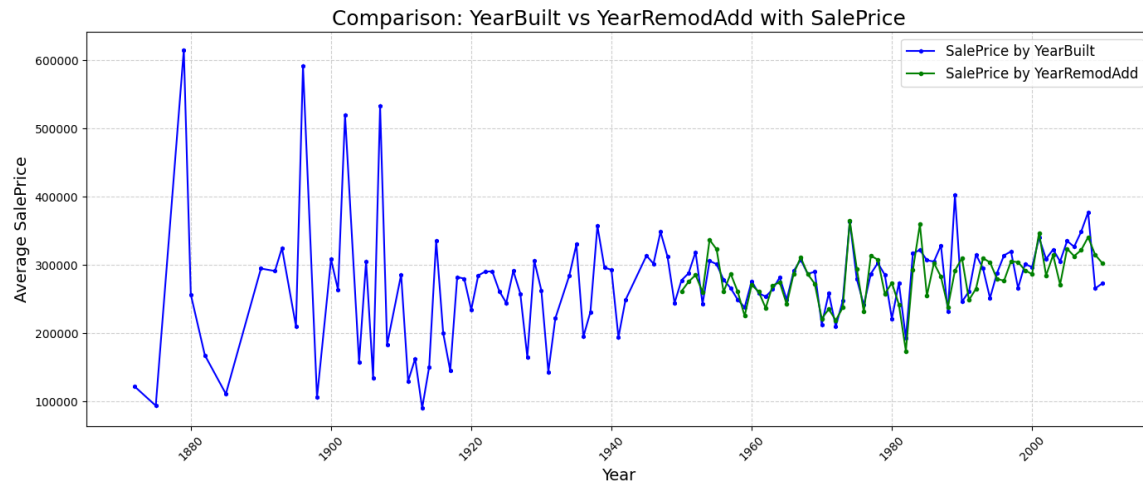
The scatterplots for TotalBsmtSF reveal that basement size exerts a meaningful influence on sale price. Homes with zero basement space show wide price variation but mostly cluster in lower-value brackets. As basement size increases, sale price tends to rise, forming a clear upward pattern in the scatterplot. There is a dense concentration of homes with basements in the range of 800–1200 square feet, indicating this is a common and competitively priced segment. Beyond 2000 square feet, prices consistently escalate, reflecting the premium placed on larger structural space. Only a handful of homes exceed 3000 square feet of basement, and these properties rank among the highest-priced homes in the dataset.

This trend confirms that basement area is one of the strongest structural drivers of valuation. The relationship is not perfectly linear due to influences such as neighborhood quality and remodeling level, but the positive association is clear and economically meaningful: bigger basements support higher pricing, elevate perceived usable space, and contribute directly to the property's market desirability.



Remodeling & Renovation Impact:

The comparison between YearBuilt and YearRemodAdd shows that remodeling has a more consistent, stronger correlation with sale price than the original construction year. Homes remodeled after 1990, especially after 2000, display major price uplift. Renovation resets the effective age of the property in buyers' eyes, allowing older homes to compete with modern builds. This behavior highlights remodeling as a strategic investment that significantly improves resale value.



7. Conclusion

This updated analysis confirms that house prices are influenced most strongly by location, structural space, and modernization, while basic condition ratings and some raw numerical features show weaker direct effects. Basement size, zoning classification, lot configuration, and remodeling year emerge as the dominant value drivers in this dataset. The presence of a larger basement is particularly impactful, showing a clear upward price trend and strong buyer preference for usable structural space.

The dataset, after cleaning, transformation, and feature engineering, is now fully optimized for advanced machine learning models such as Adaboost and XGBoost. The insights and visuals presented form a comprehensive foundation for predictive modeling and strategic decision-making in real estate valuation.